

Parth Patel

Mechanical Engineering Student | Aerospace & Mechanical Design

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Experience

Manufacturing Engineering Intern

May – Aug. 2026

Ben Machine Products

- Optimized CNC machining run times across 3-axis and 4-axis production jobs, reducing total machining time by approximately **150 hours** and generating roughly **\$7,500 in manufacturing cost savings**.
- Operated and supported 3-axis and 4-axis CNC machining processes for precision aerospace and industrial components across multiple active production jobs.
- Deburred and finished **250+ machined components**, completing work in-house and contributing to approximately **\$3,500 in deburring cost savings**.
- Read and interpreted engineering drawings, work instructions, tolerances, and dimensional specifications to support accurate machining, inspection, and production.
- Conducted dimensional and visual inspections using precision measuring equipment to verify dimensions, surface finish, and overall part quality.
- Assisted with First Article Inspection reports and manufacturing process documentation for precision-machined components.

Warehouse Associate

Jun. – Sep. 2024

Franklin Empire

- Received, inspected, organized, and prepared electrical inventory while maintaining accurate product locations and reducing inventory discrepancies.
- Supported product picking, put-away, and order fulfillment while coordinating with team members and supervisors to meet daily shipping deadlines.
- Organized incoming and outgoing materials to improve product retrieval and maintain efficient warehouse workflow.

Educational Support Assistant

Aug. – Sep. 2022

Roberta Bondar Public School

- Provided one-on-one academic support, assisted teachers with instructional materials, and maintained student attendance and progress records.

Projects

2-Axis Thrust Vector Control System

2026

CAD, Mechanical Design, Arduino, 3D Printing

- Designed and prototyped a compact **2-axis thrust vector control mechanism** using two independently actuated servo-driven rotational axes for pitch and yaw control.
- Developed the full mechanical assembly in CAD through multiple design iterations focused on gimbal geometry, servo positioning, linkage alignment, structural clearance, and nozzle range of motion.
- Designed and manufactured custom gimbal, mounting, linkage, and structural components using 3D printing for rapid physical prototyping and fit testing.
- Integrated **2 servo motors** with an Arduino-based control system and programmed directional inputs for independent actuation of both rotational axes.
- Evaluated servo placement and linkage geometry through physical assembly testing, identifying mechanical interference and constrained-motion issues within the nested gimbal mechanism.
- Applied an iterative design and prototyping process involving CAD development, additive manufacturing, physical assembly, testing, and successive design refinement.

Summary

Second-year Mechanical Engineering student at McMaster University with hands-on experience in aerospace manufacturing, CNC machining, CAD, additive manufacturing, Arduino prototyping, and mechanical design.

Interested in aerospace, propulsion, and high-performance mechanical systems. Aspiring to build a career developing aerospace hardware and advanced mechanical technologies.

Education

McMaster University

B.Eng., Mechanical Engineering

Hamilton, ON

2025 – 2029

Awards & Leadership

2nd Place – Healthy Living in Space

National Space Society Space Settlement Design Competition

Honourable Mention

National Space Society Space Settlement Design Competition

Designed structural and material systems for a large-scale torus-shaped orbital settlement around Titan as part of an international space-settlement design competition.

Co-Founder – Advancing Future Innovators (AFI)

Co-founded a STEM-focused nonprofit delivering hands-on engineering and technology workshops to younger students.

Helped develop technical activities, coordinate volunteers, and introduce students to engineering and practical problem solving.

Technical Skills

CAD & Design

Autodesk Inventor, SolidWorks, Engineering Drawings

Manufacturing

3-Axis CNC, 4-Axis CNC, FDM 3D Printing, Deburring, Finishing

Inspection

Precision Measurement, Dimensional Inspection, First Article Inspection Support

Electronics

Arduino, Servo Motors, Basic Circuit Integration

Programming

Python, Arduino Programming

Engineering

Mechanical Assembly, Prototype Testing, Design Iteration, Rapid Prototyping